

Bacon and the scientific method



Frontispiece of Bacon's great work, the *Novum Organum*.

Part of the legacy of the Islamic 'Golden Age' in European thought was a wealth of scientific scholarship, based on Aristotelian ideas of observation, analysis and classification, but with a distinctively Islamic tradition of experimentation, especially in fields such as medicine and alchemy. During the Renaissance, European scientists adopted these principles, leading to significant advances in astronomy, mathematics and the biological sciences. A turning point came, however, at the beginning of the 17th century with

Francis Bacon's *Novum Organum*. Rather than simply continuing the tradition, this English philosopher examined the method of scientific enquiry itself, and proposed a more systematic approach. Bacon advocated a process of observation, amassing and analysis of data, formation of a hypothesis and confirmation through critical experiments. This process of induction – deriving a general rule from a series of instances, forms the basis of modern scientific practice, and Bacon's insistence on evidence also influenced the British empiricist movement (see [Empiricism](#)).